

1	Report ID: 3d1020cb-6fa0-4e70-8983-d851c527843e
2	Source File: corrosion.jpg
3	Generated: 2026-04-24 07:30:05Z
4	Report Title: Image Analysis Report
5	Analysis Date: 2026-04-24
6	Analyst: AI System v2.1
7	Image Dimensions: 1024x768
8	File Size: 2.1 MB
9	Metadata: EXIF data present
10	Analysis Status: Complete
11	Report Version: 1.0
12	Classification: Severe Corrosion
13	Keywords: Rust, Corrosion, Metal Degradation
14	Tags: #corrosion #rust #metal #inspection
15	Notes: Significant localized corrosion observed.
16	Recommendations: Immediate repair required.
17	Next Steps: Visual inspection of surrounding area.
18	Follow-up: Re-inspection in 30 days.
19	Archive: Report and image saved to database.
20	End of Report

Image Analysis Report

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Executive Summary

The image exhibits severe localized corrosion and significant coating failure, particularly concentrated along a horizontal seam. Given the visible metal delamination and scaling, the severity is high, posing a risk to structural integrity or pressure containment.

Key Findings

#	Finding
1	Severe localized corrosion in the center with visible material delamination and 'rust scabbing'.
2	Preferential corrosion attack along a horizontal feature, likely a weld seam or lap joint.
3	Widespread coating breakdown leading to secondary atmospheric corrosion and rust staining.
4	Presence of heavy, dark iron oxide scale suggests a prolonged period of moisture retention and oxidation.
5	Surface pitting is suspected under the central scaled area but cannot be quantified without cleaning.
6	Visible vertical rust streaks indicating active corrosion run-off from the primary site.

Localized detections (VLM, approximate)

- Severe Scaling and Metal Loss (conf. 95%): x=0.350, y=0.150, w=0.380, h=0.650
- Seam Corrosion (conf. 88%): x=0.000, y=0.450, w=1.000, h=0.120

Recommendations

#	Recommendation
1	Perform immediate Ultrasonic Thickness (UT) gauging at the center of the defect to assess remaining wall thickness.
2	Abrasive blast clean the area to SSPC-SP 10 (Near-White Metal) to fully expose the extent of metal loss.
3	Evaluate for a weld-on repair patch or structural reinforcement if wall loss exceeds the allowable limit (typically 12.5-20% depending on code).
4	Perform Dye Penetrant Inspection (DPI) or Magnetic Particle Inspection (MPI) on the horizontal weld seam to check for surface-breaking cracks.
5	Apply a high-build epoxy coating system specifically formulated for salt/industrial environments after surface preparation.
6	Establish a localized monitoring point and re-inspect within 6 months to ensure the repair/coating remains intact.
7	Ensure all personnel maintain a safe distance until the integrity of the component (if pressurized) is verified.